Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A rover travels straight *east* at a steady **6 m/s**. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.

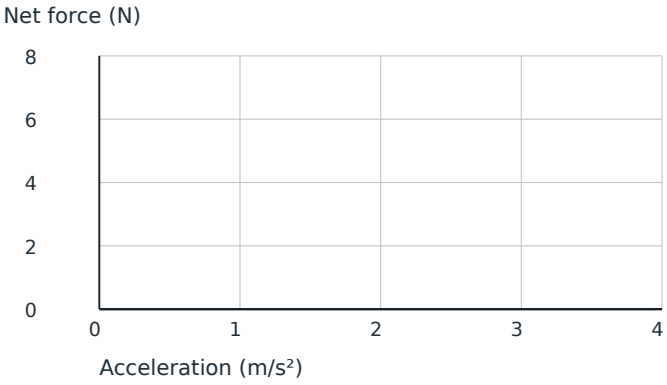
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = F_{net}/M$; $F_{drive} = m_h g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A **2 kg** cart has net forces **2** and **6 N** in two trials. Using a=F/m gives accelerations **1** and **3 m/s²**. Tripling net force triples acceleration.

Your turn: 1. **Draw a force-versus-acceleration graph** for these trials.
2. **Explain the meaning of the** slope.



Marzano check - circle: ① ② ③

03 Read a proportional pattern

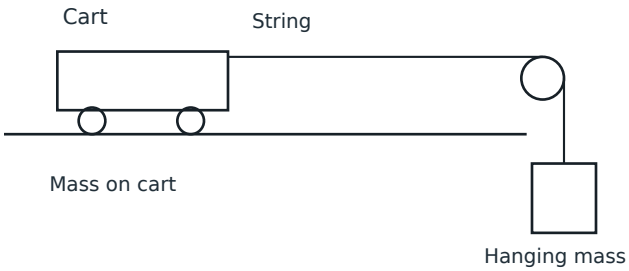
1. Use the supplied example data for the same **2 kg** system.
2. **What happens** to acceleration when force doubles?
3. **Predict a** at **8 N**.

Net force (N)	a(m/s²)
2	1
4	2
6	3

Marzano check - circle: ① ② ③

04 Keep mass constant

A student increases the hanging mass by adding a mass from the supply box.
1. Use the setup diagram to **name the two quantities this** changes.
2. **Describe how** to change driving force while keeping total moving mass constant.



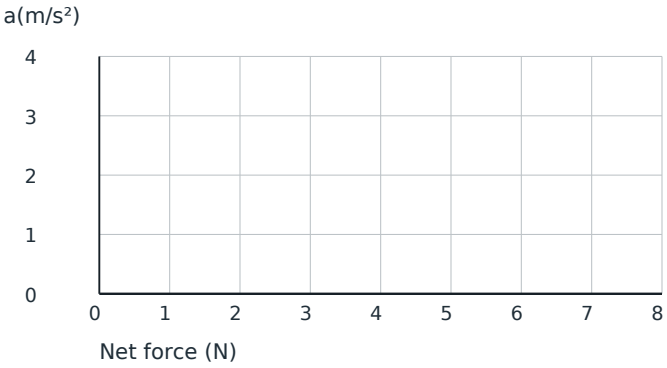
Marzano check - circle: ① ② ③

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05 Graph the pattern

Plot the supplied example data from **3**. **Label both axes** and **draw a trend line**. If you collect your own data, put those in a separate table and **label them measured**.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

With a fixed net force, a **3 kg** system accelerates at **10 m/s²**. **Predict acceleration** for a **6 kg** system. **State a control needed** for this comparison.
Use the table to keep the controlled and changed quantities distinct.

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

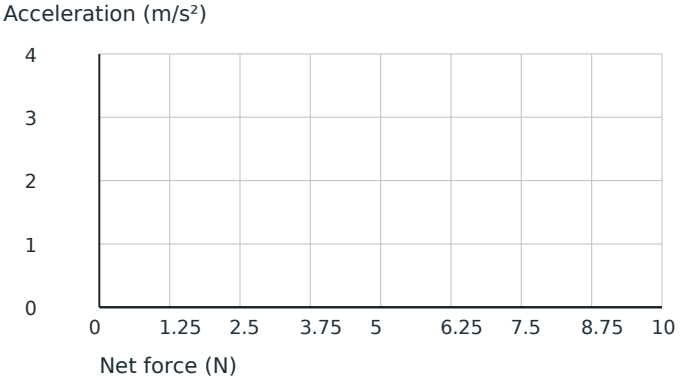
07 Exit check

A fixed-mass system accelerates at **2 m/s²** under **4 N** net force.
1. **Predict its acceleration under 6 N** net force.
2. Explain with a comparison.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Design three net-force settings that give accelerations **1**, **2**, **3 m/s²** for a **3 kg** cart.
2. **Explain what must stay fixed**.



Marzano check - circle: ① ② ③